Matched-Filter Noise Recovery and a Twelve-Member Sound-Event Ensemble for Indic Noise Event Detection

Saifuddin
Team syehsyoh (independent participant); Senior Digital Transformation (AI & Automation) Expert
Code, heads and submissions: https://github.com/syehsyoh-dokling/NoiseDetection

***Abstract*—Datathon@IndoML 2026 Track 1 asks for the onset and offset of every noise event in Vaani field recordings, scored by an event-based F1 (both boundaries within 20 % of the reference duration, at least 50 ms) plus a per-clip segment Dice. Our final submission (Codabench 994586) placed 5th of 82 leaderboard entries with 1.600895 (F1 0.780264, Dice 0.820631; maximum 2.0). The synthetic clips appear to be clean speech with noise pasted from a finite library; mixture minus the released clean reference recovers the pasted waveforms, and a normalised cross-correlation (NCC) matched filter over 1,198 such templates finds library waveforms in the test clips. Clips without a library hit go to a frame-level sound-event detector (SED): a weighted mean of twelve BiGRU-head members over frozen or fine-tuned Whisper, AST, BEATs and Dasheng encoders, with per-event peak-relative boundaries. Training adds the gated Vaani Gold/Silver tiers and, for one member, pseudo-labels on unlabelled test audio.**

***Index Terms*—sound event detection, matched filter, ensemble learning, pseudo-labelling, Indic speech, Vaani**

I. Training Data and Preprocessing

Vaani data. The validation release holds 2,501 clips (3.51 h, 16 kHz mono; 1,806 natural, 695 synthetic with clean references; 28 states, 141 state/district pairs, 27 languages) with 5,157 events in seven categories (human non-speech 1,924; vehicle/traffic 1,281; animal 1,142; baby/child 496; phone/signal/alarm 188; singing/music 68; appliance/machine 58). The test set holds 5,517 unlabelled clips (10.81 h); no test labels were released or accessed.

Additional data (allowed by the rules [1]). Gold tier of the Vaani noise-event dataset [2], [3] (10,772 clips, 19,414 verified-timestamp events; CC BY 4.0, gated) for every member except wsA and wlP; Silver tier (unverified timestamps) for wlHS (14,529 clips, HF release) and bFS (first 12,000 clips of the Kaggle copy ayanokoji2028/vaani-noise-hf). By Vaani file stem (a unique clip id), Gold shares no clip with test or validation and no recording session (file-name prefix) with test; 13 Gold clips (0.12 %) share a session with a validation clip (effect not measured). Silver extractions exclude validation and test names (not checked by session); speaker/district overlap with test was not checked.

Test-side information. (a) The leaderboard score of each upload was the final selection criterion; Codabench [1] lists 47 of our uploads (46 scored; 995069, after the final scored upload, failed) and the first (921278, 10 Sep) is no longer listed, so $\geq$48 were made (quota 5/day, 100 total). (b) Member wlP was trained on pseudo-labels [4] that three validation-only Whisper heads (small, medium, large, weighted 1/1/2) produced on the 3,353 test clips without a library hit: transductive use of unlabelled test audio (the organisers' novelty direction names unlabelled Vaani audio, not test clips); the fold-f student only sees labels from teachers that never saw fold f.

Preprocessing and augmentation pipeline. Audio is read at 16 kHz mono; no waveform augmentation or noise mixing is used. Frozen members: encoder features are extracted once and cached (float16); per batch a random crop of 400 or 600 frames (Table I), then feature-level SpecAugment-style masking [18]: 2 time masks of width $\leq$25 frames set to zero and 10 % of feature channels dropped (rescaled by 1/0.9). Fine-tuned members: a random 10.24 s crop, then on the encoder's log-mel filterbank 2 time masks ($\leq$40 frames) and 2 frequency masks ($\leq$16 bins). Silver and pseudo-labelled clips enter with sample weights 0.4 and 0.5.

Template library. Outside the pasted spans ($\pm$150 ms) the residual (mixture minus clean reference) of a synthetic clip is zero at most samples (median of per-clip medians 0, maximum 4.4×10^{-3} at full scale 1), so each of the 2,213 labelled synthetic events cut from the residual is the pasted noise up to that residual (none under the 50 ms minimum). Templates with the same tag and length ($\pm$2 samples) and NCC $>$0.995 were merged, leaving 1,198, each with its source clips for leave-one-clip-out (LOO) evaluation.

Features and validation. On a 20 ms grid, tokens of three or four layers per encoder are averaged over frequency (AST, BEATs, Dasheng), layer-normalised per frame, averaged over layers and interpolated onto the grid (Whisper frames used directly). Targets are eight binary channels (any noise + seven categories); 24 % of natural events overlap an event of another category. 5-fold cross-validation grouped by state/district (GroupKFold; 500–501 clips, 27–29 districts per fold) mimics unseen test districts; every validation number is out-of-fold (OOF) with the test decoding, LOO library and cross-fitted null levels (two halves of the natural clips calibrate each other). Decode and matcher parameters were fitted on these OOF predictions (no nested CV); each member has one seed and seed variance was not measured.

TABLE I

THE TWELVE MEMBERS. V VALIDATION FOLDS, G GOLD, S SILVER, P PSEUDO-LABELS; W-S / W-L WHISPER-SMALL / LARGE-V3; FT = ENCODER FINE-TUNED (*PATCH EMBEDDING FROZEN); CROP IN FRAMES (FR) OR S; w = ENSEMBLE WEIGHT. SEED 0, EXCEPT WLH2/WLH3 (1/2) AND BFS (7).

Tag	Encoder	Layers	Data	Ep/bs/crop	w
wsA	W-S	3,6,9,12	V	25/16/600 fr	0.5
wlP	W-L	8,16,24,32	V+P	25/16/600 fr	3
wlH/wlH2/wlH3	W-L	8,16,24,32	V+G	8/8/400 fr	2/6/1
wlHS	W-L	8,16,24,32	V+G+S	6/8/400 fr	6
aH/bH/dH	AST/BEATs/Dasheng	4,8,12	V+G	8/8/400 fr	8/8/4
aFG	AST ft (12 blocks)	4,8,12	V+G	3/8/10.24 s	8
bFG	BEATs ft*	4,8,12	V+G	3/8/10.24 s	8
bFS	BEATs ft*, no hold-out	4,8,12	all V+G+S	2/8/10.24 s	4

TABLE II

LEADERBOARD (LB) LINEAGE: 10 OF OUR ≥48 UPLOADS. OOF / ΔOOF: OUT-OF-FOLD SCORE / GAIN OVER THE PREVIOUS ROW (FOLDS UP OF 5), WHERE RECORDED. MED / THR: MEDIAN WINDOW / THRESHOLD OF THE HUMAN-NON-SPEECH BRANCH. Q1 = FINAL CONFIGURATION WITHOUT BFS.

Upload	Change	OOF	ΔOOF	LB F1	LB Dice	LB comb.
921278[c]	first upload, 10 Sep	–	–	0.7113	0.7478	1.4591
924220	v14: 6 Whisper + 2 AST frozen members	1.4466	–	0.7752	0.8055	1.5808
929944	+ aFG:8, aA dropped[b]	–	–	0.7739	0.8130	1.5868
932046	+ bH:8	–	–	0.7741	0.8148	1.5890
932698	+ bFG:8, dH:4	–	–	0.7752	0.8187	1.5938
933670	decode re-tune (med 11, thr 0.32)	–	+0.0024 (5)	0.7778	0.8194	1.5972
936587	med 13, thr 0.34	–	+0.0017 (4)	0.7799	0.8201	1.5999
991824	med 15	–	–	0.7802	0.8198	1.6000
991838	other-branch gap 2.0 (Q1)	1.4787	+0.0007 (4)	0.7806	0.8195	1.6002
994586	**+ bFS:4 (final)**	n/a[a]	–	**0.7803**	**0.8206**	**1.6009**

[a] bFS trains on all validation clips: no OOF; weight chosen on the leaderboard.
[b] Second AST member of v14, dropped after its artifacts were lost (v14 without it: 1.4474 OOF). [c] No longer listed; from our records. Leaderboard noise was not quantified, so the last three steps (≤0.0007) are not established gains.

II. PROPOSED METHOD

Background. CRNN frame classifiers [5], [6] are a standard approach to SED, scored with event- and segment-based metrics [7]; semi-supervised SED is a DCASE task [8]; Whisper keeps background-sound identity [9]. The synthetic half is known-signal detection, as in fingerprinting [15]. The matched filter is SNR-optimal only in white Gaussian noise [16]; here the interference is coloured, non-stationary speech and we do not prewhiten, so we use it as a normalised similarity score [17], not as an optimal detector.

Pretrained models (all public, roles in Table I). openai/whisper-small (HF rev. 973afd2) and openai/whisper-large-v3 (06f233f) [10], Apache 2.0; MIT/ast-finetuned-audioset-10-10-0.4593 (f826b80) [11], BSD 3; BEATs_iter3_plus_AS2M.pt from microsoft/unilm [12], MIT; mispeech/dasheng-base [13], Apache 2.0 (rebuild used e0b0a27, last modified 25 Oct 2026, after our training). Revisions are those used at the 28 Oct rebuild (September revisions not recorded); whisper-medium served only as a pseudo-label teacher; CLAP and CED [14] were only tried.

Matched filter. For a test clip and template T_i the matcher computes the energy-normalised cross-correlation (cosine, no mean removal) $c_i(s)$ at every sample lag s with FFTs on the GPU, keeping the top 2 peaks per template and clip (0.3 s apart, NCC > 0.2). A per-template null level ν_i is the 0.999 quantile of $\max_s c_i(s)$ over the natural validation clips (unstored values count as 0.2); with about 900 clips per cross-fitted half this is close to the maximum, a coarse estimate. Candidates with margin $c_i(s) - \nu_i \geq 0.1$ are selected greedily by margin with at most 30 % mutual overlap. In LOO mode the matcher fires on 687 of 695 synthetic and 73 natural clips.

SED members and loss. Each head is LayerNorm → linear (to 384) → GELU → two-layer BiGRU (hidden 192, dropout 0.3) → linear to 8 logits $z_{t,c}$. With frame mask m_t, sample weight u (1, Silver 0.4, pseudo 0.5) and channel weights $\lambda_0 = 1$, $\lambda_{1..7} = 0.5$, the loss is

$$\mathcal{L} = \frac{\sum_{b,t} u_b m_{b,t} \sum_c \lambda_c \,\mathrm{BCE}(z_{b,t,c}, y_{b,t,c})}{\sum_{b,t} u_b m_{b,t}}. \qquad (1)$$

Training uses AdamW [19] (wd 10^{-2}), a one-cycle schedule (peak 10^{-3}; ft encoders 2×10^{-5}, fp16 autocast, gradient checkpointing), last epoch kept. A member's test probability is the mean of its 5 fold heads (bFS: one model); the ensemble is $p = \sum_m w_m p_m / \sum_m w_m$. *Weights:* the eight v14 weights are a recipe recovered from a log (the search that produced them was not preserved); each later member entered at one value from a ladder (aFG 2–12, bH 2–16, bFG 4–8 or swap, dH 4–12, bFS 4–16; `models/fusion_*`), chosen by OOF checks and/or uploads (bFG: pooled OOF gain with ≥4/5 folds up); not all choices are recorded and the weights were never re-tuned jointly, so the mix is not an OOF optimum.

Decoding and fusion. Natural clips use two branches, $p_h = \min(p_{\text{hns}}, p_0)$ for human non-speech and $p_o = \min(\max_{c \neq \text{hns}} p_c, p_0)$ for the rest (p_0: any-noise channel), each decoded peak-relatively: a region where the median-filtered curve (15 / 7 frames) clears 0.2 is kept if its peak reaches 0.34 / 0.30, and both boundaries go where the curve crosses α times that event's own peak ($\alpha = 0.6$ / 0.5; +0.029 natural OOF F1 over one global threshold when introduced). Events under 0.1 s are dropped, gaps below 0.1 s / 2.0 s closed, and branches merged with duplicate removal at IoU 0.5. A clip with any library hit is treated as synthetic: template events plus any non-overlapping SED event (any-noise channel, threshold 0.7, median 3, minimum 0.1 s, gap 0.04 s). All values: `models/fusion_Q1_bFSadd4.json`. The system is offline and non-causal.

Implementation and compute. Rebuild: Python 3.13.6, PyTorch 2.14.0, CUDA 12.6 (original versions not recorded). Frozen heads, all aFG folds and bFG folds 0–2 ran on an RTX 4060 Laptop GPU (8 GB); bFG folds 3–4 and bFS on Kaggle GPUs (type not recorded). Retrain on Kaggle T4s (one GPU per job, log timestamps): aFG fold 61–63 min, bFG fold 55–60 min, bFS 90 min with test pass (original 75 min), about 11 T4-hours; a fine-tuned test pass 4.4–5.5 min; the five frozen passes <1.4 h; the matcher 84.6 min on the laptop GPU.

TABLE III
POOLED OOF SCORES ON VALIDATION: COMBINED (F1 / DICE). FOLD MEAN ± SD (RANGE): Q1 1.476±0.077 (1.379–1.590), NO WLP 1.474±0.075, SED ONLY 1.201±0.031. "RETRAINED FT": Q1 WITH RETRAINED AFG/BFG.

System	All	Natural	Synthetic
Q1 full	1.4787 (.689/.789)	1.1570 (.447/.711)	1.9868 (.993/.994)
no wlP	1.4759 (.687/.789)	1.1527 (.443/.710)	1.9866 (.992/.994)
retrained ft	1.4771 (.689/.789)	1.1542 (.445/.709)	1.9869 (.993/.994)
SED only	1.2005 (.450/.751)	1.1409 (.433/.708)	1.3378 (.476/.862)
Templates	0.8871 (.598/.290)	0.0691 (.042/.027)	1.9542 (.982/.972)

TABLE IV
NATURAL CLIPS, Q1 OOF: RECALL OF THE 2,944 REFERENCE EVENTS PER CATEGORY UNDER THE METRIC'S TOLERANCE (PREDICTIONS CARRY NO CATEGORY); MEDIAN REFERENCE DURATION IN S.

	human	vehicle	animal	baby	alarm	music	machine	all
n	1,254	655	594	265	110	40	26	2,944
recall	0.622	0.282	0.269	0.215	0.300	0.225	0.231	0.418
dur.	0.32	0.70	0.79	0.70	0.60	2.78	6.23	–

III. RESULTS AND CONCLUSION

Leaderboard. Table II: 1.4591 to 1.6009 (our highest listed upload). Changes up to v14 (more members, Gold data, split and peak-relative decodes) are not separable and give most of the gain; fine-tuning AST, dropping aA and adding BEATs and Dasheng add +0.0131; decoding +0.0063 (OOF-motivated, LB-confirmed, except med 15); bFS +0.0007, not established. The gap to rank 1 (1.6173) is 0.0164 (F1 0.0092, Dice 0.0072); selection over $\geq$48 uploads leaves an unquantified optimistic bias. OOF and LB differ by 0.12; 39 % of test clips have a library hit (validation: 28 % synthetic, 30 % LOO hits), and re-weighting the per-clip OOF Dice by that mix gives $0.608{\cdot}0.7105{+}0.392{\cdot}0.9943 = 0.822$ vs LB 0.8195, consistent with the mix driving the Dice part.

Components and errors. Table III: on synthetic clips the matcher alone gives 1.9542, the SED alone 1.3378; on natural clips the SED 1.1409, the matcher 0.0691, the fusion +0.016 over SED. In the 73 natural clips with a hit, 64 of 77 template events (83 %) match a reference event (1.3047 fused vs 0.9519 SED-only): library waveforms also occur in natural recordings, so test hits are not all synthetic clips, and such true matches raise the null levels (size not measured). Leaving one member out (decode fixed) lowers pooled OOF by 0.0097 (bFG), 0.0069 (dH), 0.0061 (bH), 0.0033 (wlH) (5/5 folds each) and 0.0027–0.0035 (aFG 3/5, wlP 4/5, wlH2 4/5, wlHS 3/5), but raises it by 0.0001–0.0014 for wsA, wlH3 and aH, which carry no OOF support. wlP's paired fold gain (mean +0.0026, $t \approx 1.7$) is not significant; its LB effect was not measured. In Table IV the shortest category (human non-speech) has the highest recall; the other six lie at 0.215–0.300 for median durations of 0.6–6.2 s; the 2.0 s gap closing, boundary ambiguity and scarcity are candidate causes we did not separate. Rejected variants (soft-Dice loss, stacker, CLAP/CED, fine-tuned Whisper/Dasheng, aH-free mixes) are in `models/`; their outcomes were not uniformly recorded.

Reproduction. After the final upload we lost backbones, fine-tuned checkpoints, cached test probabilities and template candidates; heads, configurations, library and scored zip survived. We retrained aFG, bFG, bFS with the same scripts and seeds on Kaggle T4s and recomputed the rest: retrained aFG/bFG give 1.4771 OOF (vs 1.4787); the rebuilt submission flags the identical 2,164 clips and, with the scored file as reference, scores 1.9994 on them and 1.9727 on the other 3,353 (1.9858 overall). It is not bit-identical (three members retrained; different GPUs, software and Dasheng revision; causes not separated); bFS (no OOF) rests on this agreement.

Conclusion. Known-signal detection plus per-event SED boundary placement gave 5th place. The matcher needs the test set to reuse the validation library; the ensemble is large, offline, non-causal and not jointly optimised; wlP is transductive; natural F1 (0.45) is the weakest part. Code, configurations, all member heads and retrained checkpoints are released (not wlP's pseudo-labels and teachers, so wlP cannot be retrained).

Ethics and data use. The clips record real speakers. We used the competition data and the gated Vaani release under their terms, redistribute no audio and attempted no speaker identification; consent and privacy rest on the providers' terms.

REFERENCES

[1] Datathon@IndoML 2026, "Track 1: Noise event detection," https://www.codabench.org/competitions/17825, accessed Oct. 2026.

[2] ARTPARK-IISc, "Vaani noise event dataset," 2026, https://huggingface.co/datasets/ARTPARK-IISc/Vaani-Noise-Event-Dataset.

[3] S. Pulikodan *et al.*, "VAANI: Capturing the language landscape for an inclusive digital India," arXiv:2603.28714, 2026.

[4] D.-H. Lee, "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in *ICML Workshop on Challenges in Representation Learning*, 2013.

[5] E. Çakır *et al.*, "Convolutional recurrent neural networks for polyphonic sound event detection," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 25, no. 6, pp. 1291–1303, 2017.

[6] A. Mesaros *et al.*, "Sound event detection: A tutorial," *IEEE Signal Process. Mag.*, vol. 38, no. 5, pp. 67–83, 2021.

[7] A. Mesaros, T. Heittola and T. Virtanen, "Metrics for polyphonic sound event detection," *Applied Sciences*, vol. 6, no. 6, 162, 2016.

[8] N. Turpault *et al.*, "Sound event detection in domestic environments with weakly labeled data and soundscape synthesis," in *Proc. DCASE Workshop*, 2019.

[9] Y. Gong *et al.*, "Whisper-AT: Noise-robust automatic speech recognizers are also strong general audio event taggers," in *Proc. Interspeech*, 2023.

[10] A. Radford *et al.*, "Robust speech recognition via large-scale weak supervision," in *Proc. ICML*, 2023.

[11] Y. Gong, Y.-A. Chung and J. Glass, "AST: Audio spectrogram transformer," in *Proc. Interspeech*, 2021.

[12] S. Chen *et al.*, "BEATs: Audio pre-training with acoustic tokenizers," in *Proc. ICML*, 2023.

[13] H. Dinkel *et al.*, "Scaling up masked audio encoder learning for general audio classification," in *Proc. Interspeech*, 2024.

[14] H. Dinkel *et al.*, "CED: Consistent ensemble distillation for audio tagging," in *Proc. ICASSP*, 2024.

[15] A. Wang, "An industrial-strength audio search algorithm," in *Proc. ISMIR*, 2003.

[16] G. L. Turin, "An introduction to matched filters," *IRE Trans. Inf. Theory*, vol. 6, no. 3, pp. 311–329, 1960.

[17] J. P. Lewis, "Fast normalized cross-correlation," in *Vision Interface*, 1995.

[18] D. S. Park *et al.*, "SpecAugment: A simple data augmentation method for automatic speech recognition," in *Proc. Interspeech*, 2019.

[19] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019.